

Ian Q. Snider

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EDUCATION

University of California, Berkeley , Berkeley, CA	<i>Expected: May 2030</i>
Ph.D. Nuclear Engineering	--/4.00
Washington University in St. Louis , St. Louis, MO	<i>May 2025</i>
B.S. Mechanical Engineering	4.00/4.00
Truman State University , Kirksville, MO	<i>Conferred: Dec 2024</i>
B.A. Physics, Mathematics Minor	3.89/4.00

EXPERIENCE

Lawrence Berkeley National Laboratory - Lab Affiliate, Berkeley, CA *August 2025 - present*
Principal Investigator: Bethany Goldblum - UC Berkeley
• Doctoral research.

Lawrence Livermore National Laboratory - Graduate Intern, Livermore, CA *June 2025 - present*
Faculty mentor: Jesse Norris - Nuclear Criticality Safety Division
• Developed CRISP (CRItical Simulation Pipeline) for parallelizing a Monte Carlo simulation code, COG. Utilized world class HPC machines Dane and Ruby with Slurm workload manager for parallelizing the 3500 ICSBEP reactor benchmarks. Used modern Python software development techniques such as Poetry dependency management, Git version control with YAML deployment pipeline, SQL database management, Jinja2 templating, and object-oriented programming.

Brookhaven National Laboratory - Nuclear Science Intern, Upton, NY *2022 - 2024*
Faculty mentor: Gustavo Nobre - National Nuclear Data Center
• Conducted nuclear data R&D, applying machine learning, random matrix theory, and Bayesian inference to improve heavy-nuclei resonance classification. Developed synthetic training datasets using Porter–Thomas distributed capture widths and implemented iterative learning workflows for enhanced model accuracy. Performed Monte Carlo neutron transport simulations and perturbative sensitivity analyses of thermal $1/v$ neutron cross sections for critical reactor benchmark.

Truman State University - Student Astronomy Researcher, Kirksville, MO *2021 - 2022*
Faculty mentor: Vayujeet Gokhale - Department of Physics
• Calculated trajectories of Starlink satellites to optimize telescope viewing plans. Researched long-exposure luminosity data corruption due to Starlink satellite interference. Developed a GUI for Truman astronomy students. Wrote and submitted a proposal for the TruScholars grant.

SKILLS

- **Coding Languages:** Python, Go, C, C++, Bash, Lua, SQL, MATLAB, Mathematica, Octave, LaTeX, Typst
- **Software:** Simulink, SolidWorks, OpenMC, COG (transport code), NJOY2016, Vim, Neovim, Pytorch, Scikit-learn, Slurm, Git, Linux, Arduino, Microsoft Office

- **Technical/Laboratory:** Technical Writing, Machining, Analog & Digital Electronics, Robotics, Embedded Electronics, Computer Clusters
- **Advanced physics/engineering coursework/lab experience:** Electricity & Magnetism, Electronics, Classical Mechanics, Quantum Physics, Mathematical Physics, Vibrations, Nuclear Physics, Thermodynamics, Fluid Mechanics, Solid Mechanics, Heat Transfer, Acoustics, Materials Science, Thermal Systems, Aerospace Propulsion, Turbojets, Ramjets, Autonomous Aerial Vehicle Control, State Estimation, and Kalman Filtering
- **Advanced mathematics coursework:** Linear Algebra, Ordinary Differential Equations, Computing Structures, Control Systems, Machine Learning, and Optimizations

ACTIVITIES

MARINER Robotics Project - Project Lead 2024 - 2025

- Collaborated with other students to develop an advanced autonomous underwater vehicle (AUV)
- Researched and developed hydrodynamic dive control
- Built a chassis and buoyancy engine
- Researched translational acoustic-RF communications (TARF) for data transmission at the water-to-air interface

WashU Climbing - Member 2024 - 2025

- Indoor & outdoor bouldering, top roping, and lead climbing

MATE ROV Robotics Competition - Mechanical Lead 2023 - 2025

- Mechanical team lead on the MATE ROV underwater robotics team
- Designed and built a vertical profiling buoyancy engine
- Designed grabbers and manipulators for the main ROV chassis

Society of Physics Students - Demo Chair 2020 - 2023

- Organize, develop, and perform physics demos
- Inform/encourage students to engage in research activities
- Weekly commitment to volunteer physics tutoring
- Wrote and proctored exams for 2022 & 2023 Science Olympiads (“Crave the Wave” and “Remote Sensing”)

Competitive Math December 2022

- Participated in the 2022 Putnam competition

Dark Sky TSU 2021 - 2022

- Group at Truman State University dedicated to light pollution education and outreach

HONORS

Kenneth L. Jerina Prize for Outstanding Dual-Degree in Mechanical Engineering April 2025

- Honored for academic achievements and engineering community contributions.

Conference Experience for Undergraduates 2023 September 2023

- Competitive research abstract award
- Invitation to present a research poster at the APS DNP Fall 2023 conference on The Big Island, HI

Conference Experience for Undergraduates 2022 August 2022

- Competitive research abstract award

- Invitation to the poster presentation at the APS DNP Fall 2022 meeting in New Orleans, LA

Sigma Pi Sigma Honor Society

May 2022

- Recognized for service and academic scholarship in physics

PUBLICATIONS

- [1] I. Snider, G. Nobre, D. Brown, and W. Fritsch, “Accuracy Correlation in Neutron Resonance Reclassification,” *Bulletin of the American Physical Society*, vol. 67, 2022.
- [2] I. Snider, G. Nobre, and D. Brown, “Resonance Capture Widths for the Bayesian Resonance Reclassifier,” in *APS Meeting Abstracts*, 2023, pp. DB3–74.
- [3] G. Nobre *et al.*, “Annual Report on NCSP Technical Support task in BNL during FY24,” 2024.

CONFERENCE PRESENTATIONS

- Lawrence Livermore National Laboratory Student Research Conference. Lawrence Livermore National Lab B511, Livermore, CA, August 7th, 2025.
- Brookhaven National Laboratory Student Research Conference. Brookhaven National Lab Bldg. 488, Upton, NY, August 9th, 2024.
- American Physical Society - Division of Nuclear Physics and Japan Physical Society joint fall meeting. Hilton Waikoloa Village, The Big Island, HI, Nov 27-Dec 1, 2023.
- Brookhaven National Laboratory Student Research Conference. Brookhaven National Lab Bldg. 488, Upton, NY, August 10th, 2023.
- Truman State University Student Research Conference. Truman State University, Kirksville, MO, April 21st, 2023.
- American Physical Society - Division of Nuclear Physics fall meeting. Hyatt Regency Hotel, New Orleans, LA, October 29-31, 2022.
- Brookhaven National Laboratory Student Research Conference. Brookhaven National Lab Bldg. 488, Upton, NY, August 11th, 2022.